



PG-DITISS Feb 2025 Syllabus and Marks Distribution

Duration: 80 classroom hrs (40hrs theory + 40 lab hrs)

Objective: To introduce the student to fundamentals of computers and networks

Evaluation method: Theory exam – 40% weightage
Lab exam – 40% weightage
Internal Assessment– 20% weightage

List of Books / Other training material

Courseware: Data Communications and Networking, Behrouz A. Forouzan, McGraw Hill Education

Session 1 & 2:

- Internetworking
- OSI model
- Ethernet
- Wireless Networking

Assignment:

- 1) Difference between UTP & STP
- 2) Write categories of cables
- 3) e in CAT5e
- 4) OSI Model

Session 3:

- Internet Protocol
- TCP/IP model

Assignment:

- 1) IP
- 2) TCP/IP Model
- 3) Write a difference between TCP & UDP.

Session 4 & 5:

- IP Subnetting & variable Length Subnet Masking

Assignment:

- 1) Subnetting
- 2) Determine the network and host part of 192.168.5.85 /24, 10.128.240.50/30 address.
- 3) Divide the network as per following requirement 192.168.1.0/24
 - a. Subnet 1 =28 hosts
 - b. Subnet 2 =52 hosts
 - c. Subnet 3 =15 hosts
 - d. Subnet4 = 5 hosts

Session 6:

- Router IOS & Security Device Manager

Assignment:

- 1) Subnet the Class C IP Address 195.1.1.0 So that you have 10 subnets each with a maximum 12 hosts on each subnet. List the Address on host 1 on subnet 0,1,2,3,10.
- 2) Divide the network to find 500 hosts in each subnet.
152.152.0.0 /16

Session 7:

- Managing an Internetworking Router

Lab Assignments:

Working with Router Booting, configuration registers, Router IOS, Telnet, Resolving Hostname, Debugging.

Session 8:

- Static Routing
- Dynamic Routing
- Routing Protocols

Theory Assignment:

- Dynamic Routing
- Static Routing
- Routing Protocols

Session 9 & 10:

- Implementing of Routing Protocols

Lab Assignments:

- Implementation of Static Routing, RIP, IGRP, EIGRP, OSPF

Session 11:

- Layer 2 switching

Theory Assignment:

- 1) Spanning Tree Protocol (STP)?
- 2) Write types of Spanning Tree Protocol (STP).
- 3) priority number of Spanning Tree Protocol (STP) 32768 Priority range: 0 – 61440
- 4) disable Spanning Tree Protocol (STP)

prevents loops in a network with redundant paths. It ensures there is only one active path between switches by blocking redundant links and dynamically adjusting if a failure occurs.



Lab Assignments:

1. Configuration of switch, STP
2. Create the following topology for STP and configure a specific switch as root bridge by changing priority number.

Session 12:

- Virtual LANs

Lab Assignments:

1. Configure, verify, and troubleshoot VLANs (normal/extended range) spanning multiple switches, inter-VLAN routing
2. Create following topology where Switch 1 is Server mode and switch 2,3,4,5 are client mode. Use network 192.168.10.0 /24
 - Transfer the Vlan info from server to all clients.
 - Assign vlan as mention in the topology.
 - PC's in having same vlan should ping each other.
3. Create following topology where Switch 1 is Server mode and switch 2,4, are client mode.
Use network 192.168.1.0 /24
 - Transfer the Vlan info from server to all clients.
 - Assign vlan as mention in the topology.
 - PC's in having same vlan should ping each other.
 - Configured switch 3 in transfer mode and transfer vlan info to switch 4

Session 13:

- Infrastructure Security

Lab Assignments:

- Configure, verify, and troubleshoot port security, describe common access layer threat mitigation, Configure, verify, and troubleshoot IPv4 and IPv6 access list for traffic filtering techniques,
- Verify ACLs using the APIC-EM Path Trace ACL analysis tool
- Configure, verify, and troubleshoot basic device hardening
- Describe device security using AAA with TACACS+ and RADIUS

Session 14:

- NAT
- IPV6
- WAN Technologies

Lab Assignments

NAT configuration, Configuring Routers with IPv6

Session: 15**Lab Assignments:**

- Configure and verify PPP and MLPPP on WAN interfaces using local authentication
- Configure, verify, and troubleshoot PPPoE client-side interfaces using local authentication, Configure, verify, and troubleshoot GRE tunnel connectivity.
- Configure and verify single-homed branch connectivity using eBGP IPv4 (limited to peering and route advertisement using Network command only)

Session 16:

- Introduction to SDN,
- Overview and Architecture of SDN
- Scalability (Data Centers, Service provider networks, ISP Automation)
- Reliability (QoS, and Service Availability)
- Consistency (Configuration management, and Access Control Violations)
- Opportunities and Challenges

Theory Assignment:

- SDN and Architecture of SDN.

Session 17:

- Virtual networking
- Use-cases (Network Access Control, Virtual Customer Edge, Datacenter Optimization)

Theory Assignment:

- Virtual networking
- Use cases of Network Access Control, Virtual Customer Edge, Datacenter Optimization

Session 18:

- Introduction to OpenFlow
- History and evolution
- Control and data plane separation
- OpenDaylight architecture overview

Lab Assignment:

- Understanding the classroom environment
- Installing OpenDaylight
- Initial Configuration
- Initial Controller Interactions

Session 19:

Getting Started with Opendaylight

- How to get started
- OpenDaylight clustering
- Model-Driven Service Abstraction Layer
- Internal datastore
- OpenFlow plugin
- OpenVSwitch concepts
- Mininet overview
- L2Switch Application

Lab Assignment:

- Configuring an OpenDaylight Cluster
- Initial Command Line Interactions
- Enabling Required Features
- Launching the User Interface
- Creating an Emulated SDN Network
- Running L2Switch

Session 20:

Getting more from Opendaylight

- OpenDaylight and AAA
- Introduction to OVSDB Virtualization
- Application Intents and Group Based Policy
- Service Function Chaining
- LISP Flow Mapping
- Virtual Tenant Networks

Lab Assignment:

- Advanced AAA integrations
- Creating Virtual Networks
- Working with policies and intents
- Initial Service Function Chaining
- OpenDaylight and OpenVSwitch

Duration: 90 Classroom hrs + 90 Lab hrs (Total: 180 Hrs)

Objective: To introduce the student to concepts of Operating Systems & Administration.

Evaluation method: Theory exam – 40% weightage
Lab exam – 40% weightage
Internal Assessment – 20% weightage

List of Books / Other training material

Courseware: Linux All-In-One for Dummies, by Emmett Dulaney (Wiley)

Reference book:

- Mastering Windows Server 2016 R2 by Brian Svidergol, Vladimir Meloski
- Windows Server 2022 Administration Fundamentals by Bekim Dauti

Note: Encourage candidates to implement lab using scripts.

Windows Operating System and Security Issue (80 Hrs - 40 Hrs Theory + 40 Hrs Lab)

Session 1 & 2:

- Overview of windows operating system
- Installation of windows operating system

Lab:

- Install, upgrade, and migrate servers and workloads
- Create, manage, and maintain images for deployment

Session 3:

- Concept of Windows Server Backup (WSB)

Lab:

- Implement Windows Server Backup (WSB)

Session 4 & 5:

- Implementation, planning and maintaining of active directory infrastructure

Lab:

- Implementing and administering Active Directory
- User accounts and groups in an Active Directory Domain
- Implementing an Active Directory Domain Services (ADDS)
- Domain Controller (DC)
- Additional Domain Controller (ADC)
- Client (Windows 10/11)
- Member Server (MS)

Session 6 & 7:

- Configuring DNS
- Implement DHCP and IPAM

Lab:

- Configuration of DNS server
- Install and configure DHCP
- Manage and maintain DHCP
- Implement and Maintain IP Address Management (IPAM)

Session 8:

- Local policies and Group Policies

Lab:

- Implement Local policies & Group policies

Session 9:

- Implement Network Connectivity and Remote Access Solutions

Lab:

- Implement network connectivity solutions
- Implement virtual private network (VPN) and Direct Access solutions

Session 10:

- Concept of IIS Web Server

Lab:

- Implement of IIS Web Server

Session 11:

- Concept of Core and Distributed Network Solutions

Lab:

- Implement IPv4 and IPv6 addressing
- Implement Distributed File System (DFS) and Branch Office solutions

Session 12:

- Concept of Windows Deployment Service (WDS)

Lab:

- Deploying Windows 10/11 using WDS

Session 13 & 14: Hyper-V & Storage Solutions

- Implement Hyper-V
- Configure disks and volumes
- Implement server storage
- Implement data deduplication

Lab:

- Install and configure Hyper-V
- Configure virtual machine (VM) setting

Session 15:

- Concept of File Server Resource Manager (FSRM)

Lab:

- Implement File Server Resource Manager (FSRM)

Session 16:

- Concept of Network Policy Server (NPS)

Lab:

- Implement Network Policy Server (NPS)

Session 17:

- Concept of Network Load Balancing (NLB)
- Troubleshooting

Lab:

- Implement of Network Load Balancing (NLB)
- Maintenance and troubleshooting
- Introduction to Microsoft Windows 10 & 11 security
- Security issues at the Active Directory level
- Evaluating and analyzing workstation security
- Securing Windows services

Session 18:

- Concept of Exchange server

Lab:

- Installing and implementing of Exchange Server.

Session 19 & 20:

- Power Shell

Lab:

- Windows Power Shell Technology Background and Overview
- Power shell Error Handling and Debugging
- Windows Administration using power shell
- Background Jobs and Remote Administration
- Windows Power Shell Tips & Re-Using Scripts and Functions

Linux Operating System and Security (100Hrs) (50 Hrs Theory + 50 hrs Lab)**Session 1 & 2:**

- Introduction to Linux
- The Linux File System
- Working with Files and Directories
- Getting Started to Linux
- Revision of basic Commands
- Access control list and chmod command
- chown and commands
- Network Commands like telnet, ftp, ssh, and sftp, finger
- Use of secondary storage devices (Like: - Hard disk, Floppy, CDROM) in Linux environment and formatting of these devices.

Lab:

- Getting Acquainted with the Linux Environment
- Use various commands in Linux system.
- (ls, cp, mv, lpr, sort, grep, cat, tac, more, head, tail, man, whatis, whereis, locate, find, diff, file, rm, mkdir, rmdir, cd, pwd, ln and ln -s, gzip and gunzip, zip and unzip, tar and its variants, zcat, cal, bc and bc -l, banner date, time, wc, touch, echo, who, finger, w, whoami, who am i, alias, unalias, touch, push, pop, jobs, ps, etc.)

Session 3 & 4:

- Installation of Linux
- The interactive Anaconda installer
- A hands-free method of installation
- Understanding the boot procedure
- Configuring the GRUB boot loader

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- The Initial RAM Disk
- Understanding run levels
- Repository & Package Management (RPM & DEB)

Lab:

- Installation of Linux, configuring Boot Loader, Yellow dog updater configuration

Session 5:

- Shutdown and Installation concepts
- Kick Start Configuration & Customization
- Deployment using Kickstart
- User administration

Lab:

- Installation of basic packages, configuring Kickstart, User Administration

Session 6:

- Network address Ipv4/Ipv6
- Using OpenSSH for network communications
- Using VNC for network communication
- Network Authentication

Lab:

- Configuring Network address, OpenSSH, VNC

Session 7:

- User & Group Management

Lab:

- User & Group Management

Session 8:

- Disk management

Lab:

- Fdisk, gdisk, LVM

Session 9:

- Network Implementation & print services

Lab:

- Network Implementation & print services

Session 10:

- Services Management
- System Configuration Files

Lab:

- Working with System configuration files, NIS Configuration

Session 11:

- Patches
- System Management
- X configuration server

Lab:

- Working with patch management, X configuration

Session 12:

- Configuration of DNS

Lab:

- Configuration of DNS

Session 13:

- Introduction to Configuring Services
- Setting up an NFS server
- Setting up an FTP server

Lab:

- Configuring NFS

Session 14:

- The Samba Server: networking with Windows system
- Configuring a DHCP server
- Configuring a DNS server

Lab:

- Configuring Samba, DHCP Server, DNS Server

Session 15:

- Configuring the Apache web server

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- Apache security & Virtual Hosting
- Configuring the Squid web proxy cache

Lab:

- Configuring Apache Web Server, Virtual hosting, Squid Proxy

Session 16:

- Understanding e-mail delivery
- Postfix Mail Server
- Dovecot: an IMAP and POP server
- squirrel Mail Web Client

Lab:

- Configuring Postfix, Dovecot, Squirrel Mail

Session 17:

- Introduction to Performance Tuning
- Maintenance and troubleshooting
- The Threat Model and Protection Methods

Session 18:

- Basic Service Security
- Logging and NTP
- BIND and DNS Security

Session 19:

- Network Authentication: LDAP & NIS
- Apache Clustering
- Load Balancer

Lab:

- Configure and implement of LDAP.
- Configure and implement of NIS.

Session 20:

- Virtual Machine management
- Virtual machine Network Configuration

Session 21 & 22:

- Introduction to BASH Command Line Interface (CLI) Error Handling
- Debugging & Redirection of scripts
- Control Structure, Loop
- Variable & String

- Conditional Statement Regular Expressions

Lab:

- Working to automate bash commands using bash scripts.
- Bash script using loops, Strings, regular expressions

Session 23:

- Automate Task Using Bash Script
- Security patches

Lab:

- Working to automate bash commands using bash scripts.

Session 24:

- Logging & Monitoring using script

Lab:

- Working to automate bash commands using bash scripts.

Session 25:

- Case Study

Duration: 34 classroom hrs + 36 lab hrs (Total: 70 Hrs)

Objective: Develop students with the ability to efficiently automate tasks, manage data, and develop custom solutions using Python and databases.

Prerequisites: Basics of Computer Fundamentals and Database

Evaluation method: Theory exam – 40% weightage
Lab exam – 40% weightage
Internal Assessment – 20% weightage

List of Books / Other training material

Courseware: Learning with Python by Allen Downey

Reference: Python for Unix and Linux System Administration

A. MySQL: (10 Theory +10 Lab)

Session 1:

- Concepts of databases, tables, and schemas.
- Syntax and examples for creating databases.
- Best practices for organizing and naming databases.
- Syntax and examples for dropping databases.

Lab Assignment:

- Create a MYSQL user.
- Create a database add tables and then add data into those tables.
- Delete/Modify a Table or Database.

Session 2 & 3:

- Writing SELECT statements to fetch data.
- Using WHERE clause for filtering data.
- Sorting data using ORDER BY.
- Basic JOIN operations to combine data from multiple tables.

Lab Assignment:

- Write SELECT statements to retrieve specific data from a database.
- Use WHERE clauses to filter data and extract relevant information.
- Apply ORDER BY clauses to sort retrieved data for analytical purposes.

Session 4 & 5:

- Using aggregate functions (SUM, COUNT, AVG).
- GROUP BY and HAVING clauses for grouped data.
- Subqueries and nested queries.

Lab Assignment:

- Utilize aggregate functions (SUM, COUNT, AVG) to perform calculations on dataset columns.
- Employ GROUP BY clauses to group data based on specified criteria.
- Use HAVING clauses to filter grouped data based on aggregate conditions.

B. Python: (24 Theory + 26 Lab)**Session 6:**

- Installing Python
- Your First Python Program
- Declaring Functions
- What's an Object?
- Indenting Code
- Testing Modules
- Native Data types

Lab Assignment:

- Try to install Python in windows and Linux (if by default not installed)
- Program on Datatypes.
- Make small program to print Hello Word
- Make a script file in Python and run that script.
- Take an input from user while running the program and print the output.
- Use for/while loop and If-else condition in program.

Session 7 & 8:

- Introducing Dictionaries
- Defining Dictionaries
- Modifying Dictionaries
- Deleting Items from Dictionaries
- Files
- Introducing Lists
- Defining Lists
- Adding Elements to Lists
- Searching Lists
- Deleting List Elements
- Using List Operators

Lab Assignment:

- Create a dictionary, print that dictionary.
- Use for/while loop while running the program.
- Insert/delete an item in dictionary.

Session 9:

- Introducing Tuples
- Declaring variables
- Referencing Variables
- Assigning Multiple Values at Once
- Formatting Strings
- Mapping Lists
- Joining Lists and Splitting Strings
- Historical Note on String Methods

Lab Assignment:

- Create a tuple, print that tuple.
- Use for/while loop while running the program.
- Insert/delete an item in tuple.

Session 10:

- Using Optional and Named Arguments
- Using type, str, dir, and Other Built-In Functions
- Object References
- Socket with Python

Lab Assignment:

- Append/insert/extend/remove/sort/reverse a list.
- Append/insert/extend/remove/sort/reverse a string.
- Use Built-in functions.

Session 11:

- Regular Expressions Using python
- Scripting in Python
- Functions and Functional Programming

Lab Assignment:

- Use regular expression to take a regular expression and search any pattern.
- Do all of these with the help of regular expression.
 - Match a single character except newline.
 - Match a word.
 - Match a Decimal word.
 - Match a single whitespace character.
 - Match any Non-whitespace character.

Session 12: (2 Theory Hrs + 4 Lab Hrs)

- Object Oriented Linux Environment
- Classes, Objects and OOPS concepts

Lab Assignment:

- Do practice of memory management questions.
- To find out Current directory of a file
- To find out all folders & files inside given folder.
- To find out absolute path and relative path.
- To check if a file or folder exists or not.
- To create/remove a Directory.
- To check the permissions.

Session 13 & 14:

- File Handling
- Directory Access Permissions
- Controls Socket
- Libraries and Functionality Programming
- Servers and Clients Arch
- Web Servers and Client scripting
- Introduction to Python web development framework

Lab Assignment:

- To read or write a file using python.
- Print everything written in a file.
- Count, how many lines are there in a file?
- Count, how many words are there in a file
- To read a web page content

Session 15:

- Libraries and Functionality Programming
- Servers and Clients Arch
- Web Servers and Client scripting

Lab Assignment:

- Create a socket between a server and a client then analyse and monitor all activity done by client.

Session 16:

- Exploit Development techniques
- Writing plugins in Python

Lab Assignment:

- With the help of socket programming try to exploit.

Session 17:

- Exploit analysis Automation Process
- Debugging basics

Duration: 50 Classrooms hrs + 60 lab hrs (Total: 110 Hrs)

Objective: To introduce the student to issues related to application security and ethical Hacking.

Prerequisites: Knowledge of database concepts

Evaluation method: Theory exam – 40% weightage
Lab exam – 40% weightage
Internal Assessment– 20% weightage

List of Books / Other training material

Courseware: Gray Hat Hacking: The Ethical Hackers Handbook by Shon Harris

Reference: Python for Unix and Linux System Administration

Application Security

(Web Application Security): (10 hrs Theory+ 14 hrs Lab)

Session 1:

- OWASP Top 10 – 2021
- Injection and Inclusion
- Cross Site Scripting
- Injection in stored procedures

Lab Assignment:

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- Give SQL queries to bypass authentication.
- Make your own cross site script and apply in any web application.

Session 2:

- Denial of Service
- Buffer Overflows and Input Validation
- Access Control

Lab Assignment:

- Try DOS attack to denial a service of any server.
- Try to do input validation to get access in vulnerable application.

Session 3:

- Web Application Security Risks
- Identifying the Application Security Risks
- Threat Risk Modelling
- Other HTTP fields
- Overview of Burp Suite features

Lab Assignment:

- Identify all risks and vulnerabilities of web applications using tools.
- Configuring Burp Suite with a web browser
- Scanning web applications for vulnerabilities using Burp Suite
- Using Burp Suite's scanner to identify common security issues

Session 4: (2 Theory Hrs + 4 Lab Hrs)

- Data Extraction
- Advanced Identification/Exploitation

Lab Assignment:

- Find vulnerability of data extraction/exploitation of a web application.
- Find vulnerabilities of HTTP Methods (1.0 & 1.1).
- Analyse web application with the help of Wireshark.

Session 5: (2 Theory Hrs + 4 Lab Hrs)

- SAST and DAST tools
- Case Study On Web Application Framework
- Use browser-jsguard Firefox add-on also to detect Malicious and Suspicious Webpages.

Lab Assignment:

- Try to do static and Dynamic application security testing by SAST and DAST tools.

Ethical Hacking (30 hrs Theory+ 36 hrs Lab)**Session 6:**

- Security Management Concepts & Principles
- Human side of Information Security's
- Threats of Information System
- Threats and attacks
- Classification of Threads and attacks

Session 7:

- Protecting Information System Security
- Security in Mobile and Wireless Computing
- Credit card frauds in mobile and wireless Computing
- Information Security Management
- Fundamentals of Information Security

Session 8:

- Cyber Crimes
- Understanding Cyber Crimes in context of Internet
- Legal aspects of open communications
- Indian Penal Law & Cyber Crimes Fraud Hacking Mischief
- International law
- Obscenity and Pornography Internet
- Introduction to Ethical Hacking
- Understanding Ethical Hacking Terminology
- Identifying Different Types of Hacking Technologies
- Understanding the Different Phase Involved in Ethical Hacking

Session 9:

- Types of Hacker Classes
- Red team , Blue Team ,Grey Team
- Ethical Hackers and Crackers
- Goals of Attackers
- Security, Functionality and Ease of Use Triangle
- Defining the Skills Required to become an Ethical Hacker

Session 10: (2 Theory Hrs + 3 Lab Hrs)

- Introduction to Ethical Hacking
- Creating a Security Evaluation Plan
- Types of Ethical Hacks
- Foot-printing and Social Engineering
- Understand How Traceroute Is Used in Foot-printing
- Define the Terms Port Scanning, Network Scanning and Vulnerability Scanning
- Understand various Scanning Methodologies
- SYN, Stealth, XMAS, NULL, IDLE and FIN Scans

Lab Assignment:

- Use NMAP, WHOIS for Reconnaissance.
- Open-source Intelligence Gathering tool:
 - Netcraft, Shadon, Google Dorks, Recon-Ng
- Use traceroute in footprinting.
- Use FOCA for finger printing.
- Use Tree Pad tool for reporting
- Using Burp Suite for Reconnaissance:
 - Spidering the web application
 - Mapping the attack surface using Burp Suite

Session 11: (2 Theory Hrs + 3 Lab Hrs)

- TCP Communication Flag Types
- Banner Grabbing and OS Fingerprinting Techniques
- How Proxy servers are used in launching an Attack?
- Http tunneling Techniques
- IP Spoofing Techniques
- Enumeration

- Password-cracking Techniques
- Cracking Windows Passwords
- Redirecting the SMB Logon to the attackers
- SMB Redirection, SMB Relay MITM Attacks and Countermeasures
- NetBIOS DOS Attacks
- DDos Attack

Lab Assignment:

- Do all these practical of following;
 - Ip spoofing
 - DNS Reconnaissance
 - DNS Enumeration
 - Host and Port Discovery (using NMAP) etc.
- Use Nessus also to find Vulnerability.

Session 12: (2 Theory Hrs + 3 Lab Hrs)

- Password-Cracking Countermeasures
- Active/Passive online Attacks
- Offline Attacks
- Keyloggers and other Spyware Technologies
- Trojans and Backdoors
- Overt and Covert Channels
- Types of Trojans
- Reverse-connecting Trojans
- Netcat Trojan
- Indications of a Trojan Attacks

Lab Assignment:

- Use KALI operating system tools to crack password

Session 13:

- Wrapping
- Trojan Construction Kit and Trojan Makers
- The countermeasure Techniques in Preventing Trojans
- Trojan-Evading techniques
- System File Verification

Session 14:

- Difference between a Virus and a Worm
- Types of Viruses
- Antivirus Evasion Techniques
- Virus Detection Methods

Session 15:

- Protocols Susceptible to Sniffing
- Active and Passive Sniffing
- ARP Poisoning

- Ethereal Capture and Display Filters
- MAC Flooding
- DNS Spoofing Techniques
- DNS hacking
- Describe Sniffing Countermeasures

Session 16:

- Types of DOS Attacks
- How DDos Attacks Work
- How BOTs/BOTNETs work
- Smurf Attacks
- SYN Flooding
- Spoofing vs Hijacking
- Types of Session Hijacking
- Steps to perform session Hijacking
- Prevention of session Hijacking

Session 17:

- Hacking Web Servers
- Web Application Vulnerabilities
- Web-Based Password Cracking Techniques
- Wireless Hacking
- WEP, WPA Authentication Mechanisms and Cracking Techniques
- Wireless Sniffers and Locating SSIDS, MAC spoofing
- Wireless hacking Techniques
- Methods used to secure Wireless Networks

Session 18: (2 Theory Hrs + 3 Lab Hrs)

- Backdoor Devices
- Distributed Dos attacks
- Biometric Spoofing
- Linux Hacking
- Linux Backdoors
- IDSs, Honeypots and Firewalls

Lab Assignment:

- Use NAGIOS, SNORT as IDS/IPS
- Try to install DMZ in your infrastructure.

Session 19: (2 Theory Hrs + 3 Lab Hrs)

- Physical Security
- Overview of Physical Security
- Need of Physical Security
- Factors Affecting Physical Security
- Penetration Testing Methodologies

Lab Assignment:

- Use Metasploit framework for penetration testing.

Session 20: Malware Reverse Engineering (2 Theory Hrs + 3 Lab Hrs)

- Types of Malware
- Malicious code Families
- Latest Trends in Malware
- Analysis of Malware

Lab Assignment:

- Do analysis of a Malware use tools of Static and Dynamic analysis.
- Case Study
- Malware Analysis Scenario – Case Study

Note: We can use **USB Pratirodh** and **AppSamvid** tools for System security.

Mobile Security: (10 hrs Theory+ 10 hrs Lab)**Session 21:**

- Introduction to Android Architecture
- Android File Structure
- Android Build Process
- Android App fundamentals
- Android Security Model
- Device Rooting

Lab Assignment

- Android Emulator setup

Session 22:

- Android Debug bridge
- Penetration Testing Tools

Lab Assignment

- Lab based on Android Debug Bridge

Session 23:

- OWASP Top 10 Mobile App vulnerabilities
- Attacks on Android Apps

Lab Assignment

- Lab based on Reverse Engineering
- Lab based on Smishing attack

Session 24:

- Web based attacks on Android devices
- Networks based attacks
- Social Engineering attacks

Lab Assignment

- Lab based on MITM Attacks
- Packet Sniffing using Wireshark
- Lab based on Phishing Attack

Session 25:

- Overview of Mobile Malware
- Android App Analysis

Lab Assignment

- Static and Dynamic Analysis of Android Apps using MobSF

Duration: 30 Classroom Hrs

Objective: To introduce the student to Concepts of Security Compliance audit.

Evaluation Method: Theory exam – 80% weightage

List of Books / Other training material

Courseware: No specific courseware for the module, faculty may share some coursematerials.

Reference Books:

- IT Audit, Control, and Security Robert R. Moeller
- IT Auditing: Using Controls to protect Information Assets, Chris Davis/TMH

Session 1 & 2:

- Introduction
- Cyber security Challenges for Organizations
- Compliance Basics
- Types of Security Audit
- Audit Decision Factors
- Security Audit Phases
- Requirements of Internal Audit Team
- Principal of Audits
- Auditor personal abilities

Session 3:

- Security Evaluation
- Evaluation Process
- Evaluation phases
- Assurance – 7 evaluation levels
- Evaluation Methodology

Session 4:

- National Institute of Standards and Technology (NIST) Overview
- Components of Framework
- Use of Framework

Session 5:

- General Data Protection Regulation (GDPR) Overview
- Types of privacy data & GDPR protect
- Key Steps to Ensure GDPR Compliance

Session 6 & 7:

- International Organization for Standardization (ISO) 2700x
- How the standard works
- History of ISO/IEC 27001
- ISO/IEC 27001:2005 domains
- Structure of Standards

Session 8:

- SOx Reports
- SOC Reports - Auditor Process Overview
- SOX COMPLIANCE AND SECURITY CONTROLS

Session 9:

- COBIT framework
- Cobit Components
- Cobit VS ITIL

Session 10:

- Health Insurance Portability and Accountability Act (HIPAA)
- HIPAA regulations
- HIPAA Titles
- HIPAA Rules

Session 11:

- Payment Card Industry Data Security Standard (PCI DSS)
- History of PCI-DSS
- Different Levels of PCI
-

Session 12:

- Centre for Internet Security (CIS) Critical Security Controls
- CIS Compliance
- CIS Controls
- CIS Benchmarks

Session 13:

- SSE-CMM Project
- History & the Need
- SSE-CMM Overview
- Using the SSE-CMM
- SSE-CMM Pilots

Session 14 & 15

- ITAA 2008
- Digital Personal Data Protection Act 2023
- Case Studies
- Internal Audit Solution for a Global Bank
- Compliance Audit of Anti-Corruption Regulations
- Can choose as per syllabus

Suggested Teaching Guideline for
Network Defense & Countermeasures PG-DITISS February 2025

Duration: 30 class room hrs + 40 lab hrs (Total: 70 Hrs)

Objective: To introduce the students to Network Defense and Countermeasures.
This includes the following:

- Network Security Concepts,
- Firewalls,
- IDS & IPS, and
- VPN

Prerequisites: OS and Network Concepts

Evaluation method:

Theory Exam: 40% weightage.

Lab Exam: 40%weightage.

Internal Assessment: 20%weightage.

List of books / Other training Material:

Courseware: Cryptography & Network Security: Principles and Practices by William Stallings.

Reference:

- Fundamentals of network and Security: Eric Maiwald/TMH

Note: each session and Lab consists of 2 Hours of duration

Session 1:

- Introduction to Information Security
- Why Information Security?
- Security: The money factor involved
- Internet Statistics - Study from a security perspective
- Vulnerability, Threat and Risk
- Qos

Lab 1:

- Install and configure an antivirus software to scan your system.
- Analyze and summarize internet security statistics from OWASP or other reliable sources.
- Conduct a vulnerability scan on your local network using OpenVAS.
- Simulate network traffic and measure QoS parameters using Wireshark.

Session 2:

- Risk Management, Exposure and Countermeasure
- Firewall
- De-militarized Zone
- Two methods of implementing firewall

Lab 2:

- Identify and categorize risks in your network using the NIST Cybersecurity Framework.
- Install and configure pfSense as a firewall on your network.
- Set up a DMZ using a virtual environment to host a web server with pfSense.
- Configure and compare packet-filtering and proxy firewalls using iptables and Squid.

Session 3:

- Packet Filtering
- Screened Host Firewall
- Stateful Inspection Firewall
- NextGen Firewall app controls
- iptables - Linux Firewall

Lab 3:

- iptables - Rule Processing
- Default Policy
- iptables - Predefined tables
- INPUT
- OUTPUT
- FORWARD

Lab 4:

- Iptables / netfilter/ Xtables-Addons
- verifying iptables / netfilter
- internal working
- iptables / netfilter - concept of targets
- DROP
- ACCEPT
- iptables - SPI Firewall
- Setting up a SPI Firewall - Standard installation
- Iptables IPv6 Rule Management

Lab 5:

- Automating iptables and scripting
- Bash Scripting to automate iptables
- Advanced iptables
- Loading of modules
- Geo IP Blocking using Xtables-Addons

Lab 6:

- Access control using iptables
- Internet sharing Using Iptables

Session 4:

- Wireshark
- Create a filters for data collection and display
- Examine real-world packet captures

Lab 7:

- Wireshark
- Examine real-world packet captures
- Iptables Port Forwarding
- Iptables use case: Fail2ban

Session 5:

- Linux Software Firewall(ClearOS Pfsense)
- Nginx & Squid Reverse Proxy
- UTM
- Server Load Balancing

Lab 8:

- Nginx & Squid Reverse Proxy
- Configure reverse proxy URL using regex
- Server Farming

Session 6 & 7:

Suggested Teaching Guideline for
Network Defense & Countermeasures PG-DITISS February 2025

- VPN – Introduction
- VPN protocols/characteristics
- VPN Functions
- Types of VPN
- SecureVPN
- Trusted VPN

Lab 9:

- OpenVPN configuration in both Linux & Windows
- Site to Site Connectivity
- Certificate & Password dependent authentication
- VPN configuration for Mobile Device

Lab 10:

- Pfsense OS - Installation and Configuration
- Installation of Pfsense OS and Basic
- Configuration

Lab 11:

- UTM Configuration
- Basic Setup of UTM

Lab 12:

- Configuration for access control and Firewall features (UTM)
- UTM VPN Configuration

Session 8:

- Hybrid VPN
- IPsec
- Tunnel mode/transport mode
- Ipv6 VPN
- Split Tunnel full tunnel VPN

Lab 13:

- VPN Configuration under Windows 2016 using RRAS
- L2TP/PPTP VPN Setup

Session 9:

- Introduction to IDS and IPS
- IDS / IPS
- Types of Attacks
- IDS
- Security Events
- Vulnerability/design/implementation

Lab 14:

- Distributed Honeynet System (Developed by C-DAC)
- Dynamically configure Honeypot
- UAC (URL Analyzer and Classifier)
- Tcpdump installation, verification and basic usage of tcpdump

Session 10 & 11:

- Attacks-traditional/distributed
- Intruder types
- Types of IDS
- IPS categories
- Defence in depth
- IDS and IPS analysis scheme
- Detection methodologies
- Principles of IDS
- Threat Hunting model

Lab 15:

- Install and compare Snort (NIDS) and OSSEC (HIDS) on your network.
- Set up Suricata as an IPS and configure rules for network intrusion prevention.

Lab 16:

- Analyze network traffic logs from Snort and Suricata to identify and respond to threats.
- Perform threat hunting using the ELK stack (Elasticsearch, Logstash, Kibana) to detect anomalies.

Session 12 & 13:

- Symptoms of attacks
- Tired architecture
- Sensors-network/host based
- Denial of services
- Dos & DDos Mitigation
- Sensor Deployment
- Agents
- Functions of IDS agents
- IDS Manager

Lab 17:

- Identify and document symptoms of various attacks using Wireshark.
- Design and implement a three-tier architecture in a virtual environment.
- Set up and configure network-based sensors with Snort and host-based sensors with OSSEC.

Lab 18:

- Simulate a DoS attack using LOIC and analyze its impact on a target system.
- Implement rate limiting and IP blacklisting using iptables to mitigate DoS/DDoS attacks.

Session 14:

- Introduction of Log Analyser
- Log
- SIEM Log Correlation and event triggering
- Introduction of SIEM
- SIEM Log Forwarding Configuration
- SIEM Log Correlation and event triggering

Lab 19:

- Snort
- Writing Basic Snort Rules
- Syslog Server
- BASE
- Configuration and deployment of SIEM

Session 15:

- Testing Snort in both Windows & Linux
- IDS architecture
- Bypassing an IDS

Lab 20:

- Testing of Snort using a simulated attack
- Nagios
- Nagios Sensor Configuration (Windows & Linux)
- Nagios with Email services
- Nagios with up/Down of Ram/Hard Disk

Duration: 18 Classroom hrs + 22 Lab hrs (Total: 40 hrs)

Objective: To introduce the student to Cyber Forensics and cyber laws.

Evaluation method: Theory exam – 40% weightage
Lab exam – 40% weightage
Internal Assessment – 20% weightage

List of Books / Other training material

Courseware: No specific courseware for the module, faculty may share some coursematerials.

Reference:

- Guide to Computer Forensics and Investigations Computer Forensics and Investigations/nelson
- Digital Forensic: For Mumbai University by Dr. Nilakshi, Dr. Dhananjay

Note: Conduct lab sessions based on case studied. Faculty can ask to students to get some casestudies and choose among that for explanation.

Session 1:
Introduction

- Overview - Computer Forensics
- Difference – Computer Crime & Un-authorized activities
- Cyber Laws
- Process of Computer Forensics (six)
- Need for forensics investigator

Session 2:

- Computer Forensics Involves
 - Preservation
 - Identification
 - Extraction
 - Documentation
 - Interpretation
- Goals of Forensics Analysis
- Types of Cyber Forensics Techniques
- Cyber forensics Procedures
- Preparation
 - What to do before the incident
 - Incident response plan

- Incident response team
- Detecting Incidents
- Chain of custody

Lab 1 & 2:

- Simulating Incident Response and Chain of Custody Handling

Session 3:

- Evidence Checkout Log
- Handling Evidence
- First Response
- Formulate/Execute Response Strategy
- Forensic duplication
- Authenticate the Evidence
- Investigation
- Common Mistakes
- Detection

Lab 3 & 4:

- Demo and lab sessions on Evidence Checkout Log, Investigation and Detection.

Session 4:

- The Initial Assessment
- Incident Notification Checklist
- Hexadecimal notation
- Practical Bits
- Slight diversion
- What is use of Hexadecimal

Session 5:

- Encoding And Encryption
- The Hex Editor
- Files
- Hashing
- Hashing DLs
- MD5 Hash collisions
- Hash Collisions
- Bit Rot

Lab 5:

- Demo and lab sessions on Encoding and Encryption.

Session 6:

- Standard Operating Procedures
- Processing Crime and Incident Scenes – Working with Windows & DOS Systems Current

Lab 6:

- Processing a Crime Scene on Windows and DOS Systems

Session 7:

- Forensics Implications
- Accreditation Standards
- Performing a Cyber Forensics Investigation
- Privacy and Cyber Forensics

Lab 7:

- Conducting a Cyber Forensics Investigation and Addressing Privacy Considerations

Session 8:

- Computer Forensics Tools:
 - Sysinternals Suite
 - FTK Forensics Tool kit
 - FTK Imager
 - OSF
 - Hex

Lab 8 & 9:

- Demo and lab sessions on Cyber Check Suit

Session 9: (2 Theory Hrs + 4 Lab Hrs)

- Concept of collecting and analyzing artefacts from systems that are in the active state
- Concept of collection and analysis of artefacts from Linux systems
- An introduction to Mobile forensics

Lab 10 & 11:

- Demo and lab sessions on Live system forensics & Linux Forensics

Duration: 30 classrooms hrs + 20 lab hrs (Total: 50 Hrs)

Objective: To introduce the student to PKI concepts.

Evaluation method: Theory exam – 40% weightage
Lab exam –40% weightage
Internal Assessment – 20% weightage

List of Books / Other training material

Courseware: Cryptography & Network Security: Principles and Practices

Reference:

- Cryptography & Network Security – Atul Kahate
- Understanding Digital Signatures: Establishing Trust Over the Internet and Other Networks – Gail Grant
- Windows Server 2008 PKI and Certificate Security [With CDROM] by Brian Kumar

Note: lab sessions are of duration 4 hrs as mentioned.

Session 1:

- Information Security
- Security Attacks & Threats

Session 2:

- Basic Encryption Concepts
- File Encryption
- Encryption Folders (Graphical/ using cipher)

Session 3:

- Cryptographic Fundamentals
- Cryptographic Ciphers (Symmetric and Asymmetric)
- Protocols (History, Usage, Key generation, Ciphering message)

Session 4:

Symmetric Key Encryption

- DES
- AES, RC5

Asymmetric Key Encryption

- RSA
- ECC

Session 5: (2 Theory Hrs + 4 Lab Hrs)

- Diffie-Hellman Key Exchange
- Attacks against encryption
- Cryptographic issues

Lab Assignments:

Using Cryptool

- Symmetric and Asymmetric key encryption and decryption
- Symmetric key encryption such as Caesar, Vernam, DES, RC4, AES, Substitution, XOR, Triple DES, etc.
- Asymmetric key encryption using RSA, ECC

Session 6:

Secure Hashing Methods

- SHA Secure Hash algorithm
- HMAC

Session 7:

PKI Fundamentals

- Digital Signature
- Digital Certificate

Session 8:

- CA
- Trust Model
- Certificate Issuance Process
- Certificate Revocation (CRL, OCSP)
- Types and Classes of Certificate

Session 9: (2 Theory Hrs + 4 Lab Hrs)

- Introduction to Aadhaar and e-Sign
- Time stamping Services

Lab:

Using XCA to create a Digital Signature

- Digitally Sign a Word document using the created certificate
- Digitally Sign a PDF document using the created certificate

Session 10: (2 Theory Hrs + 4 Lab Hrs)

Public Key Cryptography Standards

- PKCS
- FIPS 140-2

Lab:

Using XCA to create a Digital certificate

- Create a digital certificate using XCA tool
 - Create CA using XCA first and then CA will issue certificate to website using XCA.
 - a. Use certificate for host website (<https://www.ditiss.local>)
 - b. Import it client browser to remove the self-signed certificate warning.

Session 11:

- Strong Authentication
- Single Factor and Multi-factor authentication
- Single Sign-on Solutions
- Open-ID and OAUTH
- Graphical Passwords

Session 12:

- Authentication Protocols
- FIDO Authentication
- Zero Trust Architecture

Session 13: (2 Theory Hrs + 4 Lab Hrs)

Securing Websites and Emails

- SSL
- TLS
- PGP and S/MIME

Lab:

Using OpenSSL to create

- Creating self-signed Digital Certificates using OpenSSL.
- Creating digital certificate of Hierarchical Trust Model for Intranet
 - Setup a PKI infrastructure using openssl as following:

>> Root CA (on a Debian OS - rtca.pgditiss.local)

>> Sub CA (on a Debian OS - sbca.pgditiss.local)

Setup a HTTPS website (https://www.pgditiss.local) on Apache and access the same from the windows base machine.

Note:

>> The Certificate for www.pgditiss.local to be issued by sbca.pgditiss.local

>> Configure DNS / Name Resolution as required.

Session 14 & 15: (4 Theory Hrs + 4 Lab Hrs)

- IT Act
- LDAP or Active Directory
- Introduction to Blockchain

Lab

- Digitally Sign and Encrypt Email using the created certificate through Email Clients like Thunderbird or Windows Mail / Outlook.

Duration: 46 Classroom hours + 44 Lab hours (90 Hrs)

Objective: To introduce the student to Concepts of ITIL and Data center management.

Evaluation Method: Theory exam – 40% weightage
Lab exam - 40% weightage
Internal assessment – 20% weightage

List of Books / Other training material

Courseware: Cloud Computing Black Book, Kogent Learning, Kailash , (Wiley),2024,

A. ITIM (14 Theory Hrs + 2 Lab Hrs)

a. Introduction to ITIL (Each Session from 1 to 5 includes 2 Hrs of Theory only)

Session 1:

Service Strategy

- General strategy
- Competition and market space
- Service provider types
- Service management as a strategic asset
- Key process activities,
- Financial management
- Service portfolio management
- Demand management
- Key roles and responsibilities of staff

Session 2:

Service Design

- Design of architecture, processes, policies, documentation, and allowing for future business requirements
- Service Design Package (SDP)
- Service catalog management
- Service level management
- Designing for capacity management
- IT service continuity
- Information Security

Session 3:

Service Transition

- Service Asset and Configuration Management
- Transition Planning and Support
- Release and deployment management
- Change Management
- Knowledge Management

- key roles of staff

Session 4:

Service Operation

- Balancing conflicting goals (e.g. reliability Vs cost etc)
- Event management
- Incident management
- Problem management
- Event fulfillment
- Asset management
- Service desk
- Technical and application management
- Key roles and responsibilities for staff

Session 5:

Continual Service Improvement (CSI)

- Training and awareness
- Ongoing scheduling
- Roles created
- Ownership assigned
- Activities identified to be successful

b. Data Center Management:

Session 6: (2 Theory Hrs)

Introduction

- Data Center Architecture, Requirements & prerequisites
- Required Physical Area for Equipment and Unoccupied Space
- Required power to run all the devices
- Required cooling and HVAC
- Required weight
- Required Network bandwidth
- Budget Constraints
- Selecting a Geographic Location
- Safe from Natural hazards & Manmade disasters
- Availability of local technical talent
- Abundant and Inexpensive Utilities Such as Power and Water
- Selecting and Existing building
- Characteristics of an Outstanding Design
- Guidelines for Planning a Data Center
- Data Centre structures
- Raised Floor Design and Deployment
- Design and Plan against Vandalism

Session 7: Infrastructure in a data center (2 Theory Hrs + 2 Lab Hrs)

- Modular Cabling Design
- Points of Distribution
- ISP Network Infrastructure and WAN Links
- Network Operations Center and Monitoring
- Data center physical security, Logical security, and cleaning
- Reasons for data center Consolidation
- Consolidation opportunities
- Datacenter Servers
- Server Capacity Planning
- Disaster Recovery
- Data Center Security Guidelines
- Internet Security Guidelines
- Internet security
- Source Security Issues
- Best Practices for System Administration
- System Administration Work Automation

B. DevOps (32 Theory Hrs + 42 Lab Hrs)**Session 8:**

- Introduction of virtualization
- Virtualization types: type1, type2
- Virtualization, Hardware Virtualization, Para-Virtualization, Cloning, Snapshot, and Template
- Operating System Virtualization
- Cluster Architecture
- Cluster Requirements

Assignment –Lab

- Create and configure VM using VBox
- Deploy code on VM

Session 9: Storage Area Network

- Configuring a SAN (FreeNAS)
- Using SAN for high availability
- ZFS Volume Configuration
- IP-Based Storage Communication
- Object Storage services

Lab Assignment:

- Download and install FreeNAS (TrueNAS CORE) in a virtual machine using

- VirtualBox/VMware Workstation. Configure a storage pool and create a ZFS dataset to be used as a SAN.
- Set up two instances of FreeNAS in virtual machines using VirtualBox/VMware Workstation. Configure iSCSI targets on both instances and set up multipath I/O (MPIO) host machine to simulate high availability.
 - Within the FreeNAS virtual machine, create a ZFS pool using multiple virtual disks, add datasets, and configure snapshots and replication. Use the FreeNAS web interface to manage the ZFS volumes.
 - Configure NFS and CIFS/SMB shares on the FreeNAS virtual machine. Use host PC to connect to these shares, map network drives, and transfer files.
 - Install MinIO, an open-source object storage server, within a virtual machine. Set up a bucket using the MinIO web interface/command-line tool, upload objects, and configure access policies.

Session 10: (2 Theory Hrs + 4 Lab Hrs)

- Introduction to Cloud
- Cloud Computing, Cloud SPI Model, Cloud Computing Deployment Model(Public, Private, and Hybrid), Cloud Security (SLA and IAM).
- Cloud architecture
- Service models: IaaS, PaaS, SaaS
- Services provided by Cloud (Compute, Database, Developer Tools, Storage,Media, Mobile, Web, Security, Integration, etc.)
- Cloud development best practices
- Introduction to Openstack
- HCI & Its Comparison to cloud
- SDN

Assignment –Lab

- Exploring various services provided by cloud providers like App Services, Web apps, API Apps, Search, Database Servers on VMs, VM Scale Sets, Bot Services, and other cloud applications.
- HCI Mandatory & Optional
- Virtual Network configuration using SDN

Session 11 & 12:

- Cloud API integration
- DC/DR Migration
- DC/DR Storage Synchronization
- Bootstrapping Chef/puppet Server
- Migration of physical servers to clouds

Lab Assignment:

- Cloud API Integration for DC/DR Setup
- Automated Migration and Storage Synchronization
- Configuring Chef/Puppet for Cloud Migration
- Physical to Cloud Server Migration
- Implementing High Availability with Cloud API and Configuration Management

Session 13: (2 Theory Hrs + 4 Lab Hrs)

- Centralized logging
- Nagios
- Prometheus Next Gen NMS
- Identifying bottlenecks
- Auto-scaling Auto-rebuilding cloud instances
- Updating servers without downtime
- Auto-healing
- Cloud Enable Data Center Case Study

Lab Assignment:

- Configuring Nagios on Linux server and adding Linux client for monitoring.
- Configuring Nagios on Linux server and adding Windows client for monitoring.

Session 14:

- Agile
- Agile methodologies: Scrum and Kanban
- Lean
- Implementation of Lean
- Lean and Agile in DevOps

Lab Assignment:

- Agile Fundamentals
- Agile Methodologies - Scrum
- Lean Implementation
- Lean and Agile in DevOps

Session 15 & 16:

- Introduction to DevOps
- DevOps ecosystem
- DevOps phases
- CAMS model, Kaizen, Immutable deployment
- CI/CD Pipelines
- IAM, LXC, Docker, KVM

Lab Assignment:

- Create an httpd container running on port 8000. Name the container as web5. Do not map any directory. Create an “index.html” file & copy this file inside the container and check if the website displays your webpage. Again, modify the “index.html” file and copy it inside the container, and check if the website is updated or not.
- Create a C program(exe). Test the program on-base machine. Then create an image using Dockerfile to execute this exe program. Give image name as “Name: v1”.
- Try the above with a program that requires user input.

Session 17 & 18: Github and Jenkins

- Introduction
 - Introduction to Git
 - Core Concepts
 - Going Command Line
- Basic Git workflow with Github
 - Welcome to GitHub
 - Setup the Project Folder
 - Git Configuration (User Name and Email)
 - Copy the Repository from GitHub to Your Local Computer (git clone)
 - The First Commit
 - Publishing Changes back to GitHub (push)
- Introduction of Jenkins
- CI/CD pipeline using Jenkins.

Lab Assignment:

- Create a C program to take input from the user and test it. Then create an image using Dockerfile. Test image by the running container. Then push the image to the docker hub repository. Then also export the image in a “cprog-back.tar” file.
- Create a CI/CD pipeline using Jenkins, Docker, and GitHub to automate the web by using the NGINX image.
- Create a CI/CD pipeline using Jenkins, Docker, and GitHub to automate the web by using the httpd image.

Session 19: (2 Theory Hrs + 4 Lab Hrs)

- Introduction to AWS
- Services provided by AWS: EC2, Lambda, S3
- Introduction Virtual Private Cloud (VPC) Setup

Assignment –Lab

- Create AWS EC2 instance
- Create AWS Lambda
- Create AWS S3 bucket

- Create AWS VPC
- Create a new VPC by the name ditiss-lab with public & private subnet. Assign network address as 172.20.0.0/16. Assign 172.20.5.0/24 to the public and 172.20.10.0/24 to the private subnet. Create one instance and connect to a private subnet. Create one instance and connect to the public subnet. Install httpd on the private instance and check with curl from the public connected instance.

Session 20 & 21:

- Version Control system
- Infrastructure as Code
- Containerization with Docker
- Container Orchestration: Kubernetes, Dockerswam
- Micro Service Deployment

Lab Assignments:

- Create a new docker image using NGINX. Push it to docker hub. Create a service on Dockerswarm. Keep initial replicas as 1 and then increase the replicas to 10. Reduce the replicas to 2. Create a new image by modifying "index.html". Push to docker hub and then update the service for a new image.
- Configure a Linux as a Master node using Kubernetes and make 2 to 3 worker node clusters.

Session 22: (2 Theory Hrs + 4 Lab Hrs)

- Introduction to Ansible and Configuration Management
- Setting Up Ansible Environment
- Ansible Playbooks and YAML Basics
- Managing Ansible Inventory
- Ansible Roles and Reusability

Lab Assignments:

- Setting Up Ansible Environment
 - Install Ansible on a local machine or a virtual machine.
 - Configure SSH keys for passwordless authentication to a target node.
 - Verify the setup by running the ansible -m ping all command.
- Writing and Executing Ansible Playbooks
 - Create a simple Ansible playbook to install and start a web server (e.g., Apache or Nginx) on a target node.
 - Use the playbook to ensure the web server is running and serving a default webpage.
 - Verify the setup by accessing the webpage from a browser.
- Managing Ansible Inventory and Roles
 - Create a static inventory file listing multiple target nodes.

- Develop a role to install and configure a database server (e.g., MySQL or PostgreSQL).
- Apply the role using an Ansible playbook to deploy the database server on one of the target nodes listed in the inventory.

Session 23: (2 Theory Hrs + 4 Lab Hrs)

- Introduction to Infrastructure as Code (IaC) and Terraform
- Setting Up the Terraform Environment
- Writing and Organizing Terraform Configuration Files
- Terraform State Management
- Terraform Modules and Reusability

Lab Assignments:

- Setting Up the Terraform Environment
 - Install Terraform on a local machine.
 - Configure AWS credentials for Terraform to interact with AWS services.
 - Create and apply a basic Terraform configuration to provision an EC2 instance in AWS.
- Writing and Organizing Terraform Configuration Files
 - Write a Terraform configuration file to provision a VPC, subnet, and an EC2 instance within the subnet.
 - Use variables and outputs to make the configuration more dynamic and reusable.
 - Apply the configuration and verify the resources are created in AWS.
- Terraform State Management and Modules
 - Create a Terraform module for provisioning an EC2 instance with a security group.
 - Use the module in a Terraform configuration to provision multiple EC2 instances in different subnets.
 - Manage the Terraform state using a remote backend (e.g., AWS S3) to enable collaboration and state sharing.

Duration: 90 Hours

Objective: To reinforce knowledge of general Aptitude & English

Prerequisites: Knowledge of Mathematics & English.

Evaluation method: Theory exam– 80% weightage
Internal Assessment– 20% weightage

List of Books / Other training material

Reference:

1. Quicker math by M. Tyra (BSC publication co. Pvt. Ltd)
2. Quantitative Aptitude by RS Aggarwal
3. Verbal & Non-Verbal Reasoning: RS Aggarwal
4. Quantitative Aptitude - Quantum CAT: Sarvesh K Verma
5. High School English Grammar & Composition Revised Edition Wren, Martin / S. Chand Publisher
6. How to prepare GRE by Barron's / galgotia publications pvt. Ltd
7. Oxford Guide to English Grammar 01 Edition John Eastwood / Oxford University Press
Website to refer: www.indiabix.com
8. Business Communication by H S Mukerjee / Oxford University Press
9. Business Communication by R K Madhukar / Vikas Publishing House Pvt. Ltd.
10. Business Communication Essentials A skills-Based Approach to Vital Business English by Courtland Bovee, John Thill / Pearson
11. Effective Business Communication by Asha Kaul / Prentice Hall of India
12. Fundamental of Technical Communications by Meenakshi Raman, Sangeeta Sharma / Oxford University Press
13. English is easy by Chetan Anand Singh/ BSC publication Co. Pvt. Ltd
14. Communication Skills Publication Year 2011 Sanjay Kumar, Pushp Lata / Oxford University Press
15. Professional Communication Skills Praveen S R Bhatia / S. Chand Publishing

Note:

- Each session having 2 Hours
- General Aptitude is of 40 Hours
- Effective Communication is of 50 Hours

Part I – Aptitude

Session 1:

- Number System
 - Unit Digit
 - Last 2- digits
 - Remainder
 - Divisibility
 - Cyclicity
 - Fast Maths
 - Simplification
 - LCM- HCF

Session 2:

- Ratio and Proportion
- Partnership

Session 3:

- Percentage
- Profit and Loss

Session 4:

- Simple Interest- Compound Interest

Session 5:

- Time, Speed and Distance
- Trains

Session 6:

- Time and Work
- Wages

Session 7:

- Pipes and Cisterns
- Boats and Streams

Session 8:

- Averages
- Mixtures and Alligations

Session 9:

- Probability

Session 10:

- Permutations and Combinations

Session 11:

- Series
 - Number
 - Alphabetical
 - Repetitive

Session 12:

- Blood Relations

Session 13:

- Coding- Decoding

Session 14:

- Seating Arrangement

Session 15:

- Syllogism
- Venn Diagram

Session 16:

- Data Interpretation
- Data Sufficiency

Session 17

- Problems on Ages
- Clock & Calendar

Session 18

- Alphabetical Reasoning
- Ranking & Order

Session 19

- Direction Sense
- Puzzles

Session 20

- Statements & Arguments, Statements & Conclusions, Statements & Assumptions

Part II -Effective Communication**Session 1:**

Fundamentals of Communication

- Process of communication
- Types of communication
- Effective communication

Session 2:

The Art of Communication

- Vocabulary, spelling and grammar
- Fluency, pronunciation, intonation and accent

Practice Sessions:

Practice words, spelling, intonation and correct pronunciation

Session 3:

Personality Development

- First impressions
- Greeting
- Formal dressing & etiquettes
- Body language
- Ethics

Session 4:

Personality Development

- Developing positive attitude
- Confidence building
- Questioning techniques
- Psychometric Analysis

Practice Sessions:

Practice greeting, etiquettes and questioning

Sessions 5 & 6:

English Grammar

- Nouns
- Pronouns
- Verbs
- Adjectives
- Adverbs
- Prepositions
- Conjunctions
- Articles

Practice Sessions:

Practice sentence making

Session 7:

English Grammar

- Present Tense
- Past Tense
- Future Tense

Practice Sessions:

Practice sentence making

Session 8:

English Grammar

- Active and passive voices
- Direct and indirect speeches

Session 9:

English Grammar

- Idioms
- Synonyms & Antonyms

Practice Sessions:

Practice speaking in active & passive voices

Practice direct & indirect speaking

Practice idioms, synonyms & antonyms

Session 10:

Correct Usage of English

Session 11:

Common Mistakes in English Communication

Practice Sessions:

Practice correct English communication

Session 12:

Listening Skills

- Importance of listening
- Techniques for effective listening

Session 13:

Listening Skills

- Voice & Accent (VNA) Rounds in interviews
 - Listening to audio/video clips
 - Question-answers based on the listened audio/video clips

Practice Sessions:

Practise audio synthesis

Session 14:

Reading Skills

- Reading Comprehension
 - Practise proper accent and articulation
 - Techniques to answer questions based on comprehension

Practice Sessions:

Comprehension exercises

Session 15:

Writing Skills

- Essay writing
 - Characteristics of a good essay
 - Types of essays
 - Structure of an essay (introduction, main body, conclusion)
- Generative AI based writing

Session 16:

Writing Skills

- Letter writing
 - Types of letters
 - Parts of a letter
- Official emailing
 - Structure and etiquettes of email writing
 - Tips to write an impressive email

Practice Sessions:

Essay writing

Letter writing

Email writing

Personalized Generative AI based writing

Session 17:

Public Speaking

- Managing stage fear
- Speech design
- Informative speeches
- Speeches for special occasions (Introduction, Welcome, Felicitation, Thanks, etc)
- Personalized Generative AI based speeches
- Extempore & impromptu speeches

Practice Sessions:

Conduct various types of speeches

Preparing personalized Generative AI based speeches

Session 18:

Presentation Skills

- How to conduct effective and engaging presentations?
- Organisation & structure of presentation
- Design of slides in PPT
- Body language & voice

Practice Sessions:

Conduct presentations using PPT

Feedback of presentations

Session 19:

Group Discussions

- What is a GD?
- Skills assessed in GD
- Common mistakes
- Common GD topics

Practice Sessions:

Conduct practice GDs with video recording
Playing and analysis of GDs conducted

Session 20:

Personal Interviews

- Preparation for Interview
 - Qualities interviewers looking for
 - Getting ready for Interviews
 - Company Research
 - Overall approach
 - Just before interview

Session 21:

Personal Interviews

- Introducing yourself
 - Importance of introduction
 - Structure of introduction

Session 22:

Personal Interviews

- Elevator Pitch for effective introduction
 - Importance of Elevator Pitch
 - Structure of Elevator Pitch

Practice Sessions:

Practice introduction
Analysis and feedback on introduction
Practice elevator pitch
Analysis and feedback on elevator pitch

Session 23:

Personal Interviews

- Facing job interviews
 - Confidence
 - Body language

- Right mindset

Session 24:

Personal Interviews

- Tips for facing Interviews
 - What to do (and not do) during interviews?
 - Best practices of answering questions
 - Common mistakes of answering questions

Session 25:

- Online Interviews
- Tips for online Interviews
 - Best practices for attending online interviews
 - What to do (and not do) during online interviews?

Multiple Practice Sessions:

Practice common technical questions

Practice common HR/behavioral questions

Conduct mock interviews

Conduct online interviews

Module Code	Module Name	Hours			Marks			
		Theory	Lab	Total	CCEE	Lab	IA	Total
DITISS01	Fundamentals of Computer Networks	40	40	80	40	40	20	100
DITISS02	Concepts of Operating System and Administration	90	90	180	40	40	20	100
DITISS03	Programming Concepts	34	36	70	40	40	20	100
	IT Infrastructure Management & DevOps	46	44	90				
DITISS04	Network Defense and Countermeasures (NDC)	30	40	70	40	40	20	100
	Compliance Audit	30	-	30				
DITISS05	PKI	30	20	50	40	40	20	100
	Cyber Forensics	18	22	40				
DITISS06	Security Concepts	50	60	110	40	40	20	100
APT07	Aptitude & Effective Communications	90	-	90	40	40	20	Grade
DITISS08	Project	-	90	90	-	-	(200)	Grade
Total		458	442	900	280	280	140	600

4. Continuous Assessment

This is being done primarily by the respective faculty in the form of lab tests, assignments, quizzes etc conducted with the help of the respective course coordinators at regular intervals, as and when the portions of the modules are completed. These are basically internal exams and local to the centre. This process is further categorized into two parts.

- Lab test (40 marks): It should definitely include minimum one lab exam. It may also include assignment, viva, etc.
- Internal test (20 marks): Assignments, case studies, quiz, viva, attendance, etc. depending on the subject and the faculty.

5. Weightage of Marks

The figures shown below indicate the weightage of each component of a module in the final performance statement. The examination(s) for each module must be conducted for at least that number of marks. However, the centre may conduct evaluation for a higher number of marks, in which case the marks will be scaled down. For example, if the lab test for a module is conducted for 100 marks, the marks earned by the students will be scaled down to out of 40.

The weightage for each component will normally be:

- Theory - 40% through Centralized Course End Examinations (CCEE)
- Laboratory - 40% (Lab part of Continuous Assessment)
- Internal Test - 20% (Internals part of Continuous Assessment)

Note: Where a module does not have a practical component, the Lab component weightage will be merged with the Internal Test weightage.